

Review of Last Lecture

3.8 Convolution Theorem

3.9 Fourier Transform of Periodic Signals

3.10 Fourier Transform of Sampled Signals

3.11 Sampling Theorem

3.8 Convolution Theorem

(1) Time-domain Convolution Theorem

The time-domain convolution of two signals is equivalent to the multiplication of their spectra in the frequency domain.

$$\text{if } F_1(\omega) = \mathcal{F}[f_1(t)], \quad F_2(\omega) = \mathcal{F}[f_2(t)],$$

then

$$\mathcal{F}[f_1(t) * f_2(t)] = F_1(\omega)F_2(\omega)$$

(2) Frequency-domain Convolution Theorem

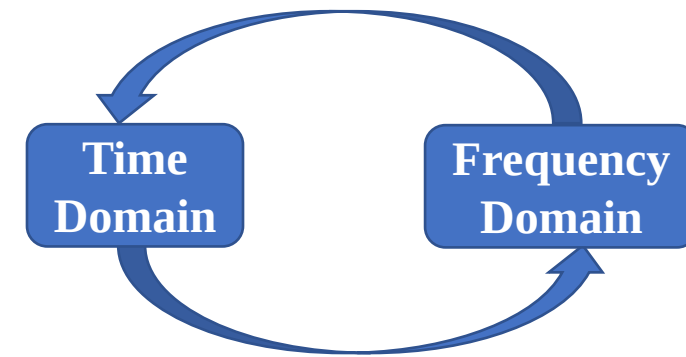
The time-domain convolution of two signals is equivalent to the multiplication of their spectra in the frequency domain.

$$\text{if } F_1(\omega) = \mathcal{F}[f_1(t)], \quad F_2(\omega) = \mathcal{F}[f_2(t)],$$

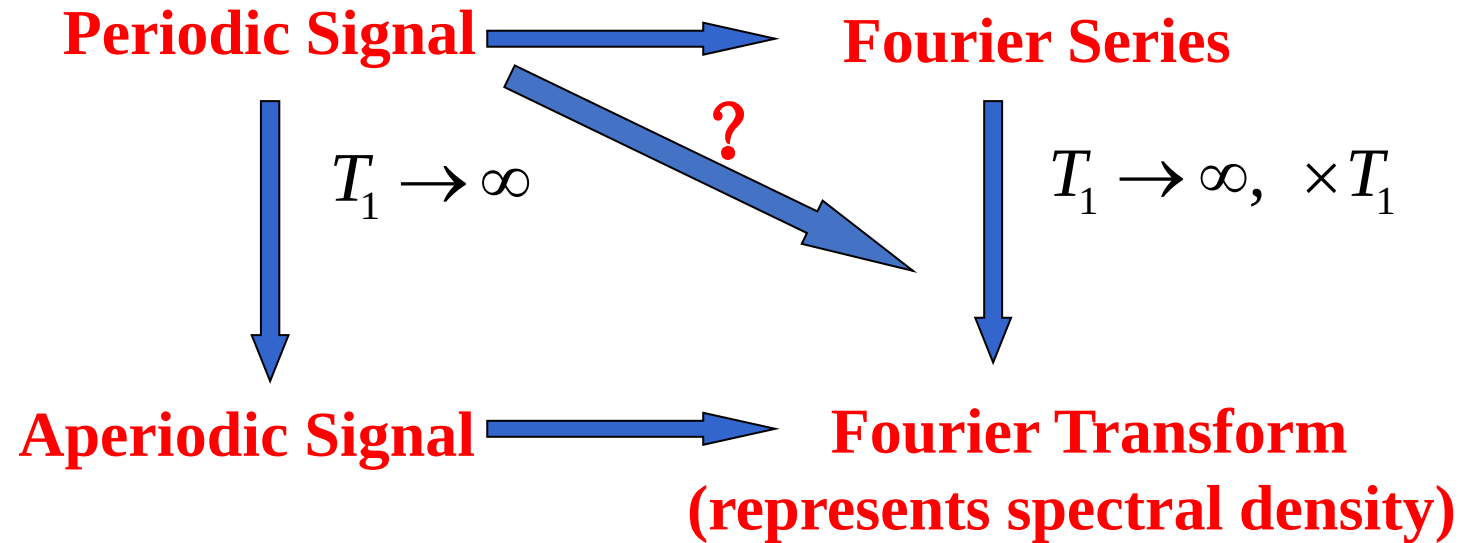
then

$$\mathcal{F}[f_1(t) \cdot f_2(t)] = \frac{1}{2\pi} F_1(\omega) * F_2(\omega)$$

$$\text{where: } F_1(\omega) * F_2(\omega) = \int_{-\infty}^{\infty} F_1(u)F_2(\omega - u)du$$



3.9 Fourier Transform of Periodic Signals



Fourier Transform of General Periodic Signals

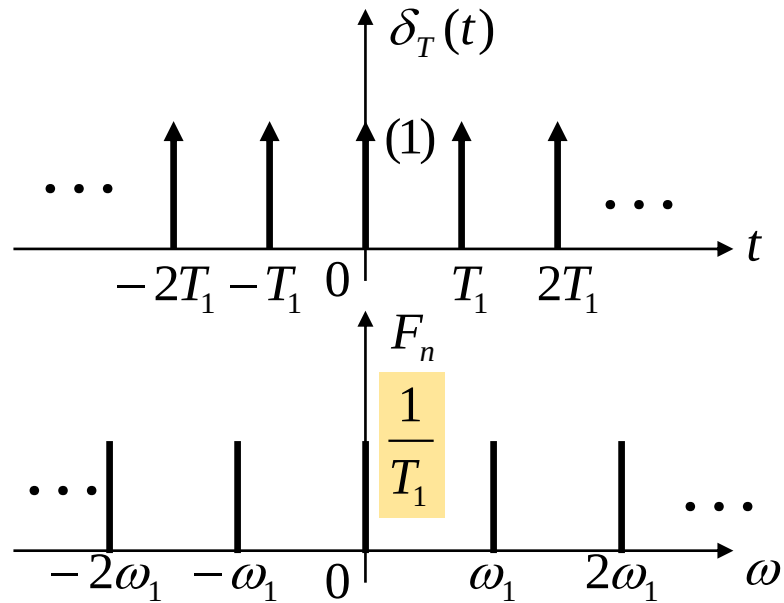
$$F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_1)$$

where: $F_n = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} f(t) e^{-jn\omega_1 t} dt$

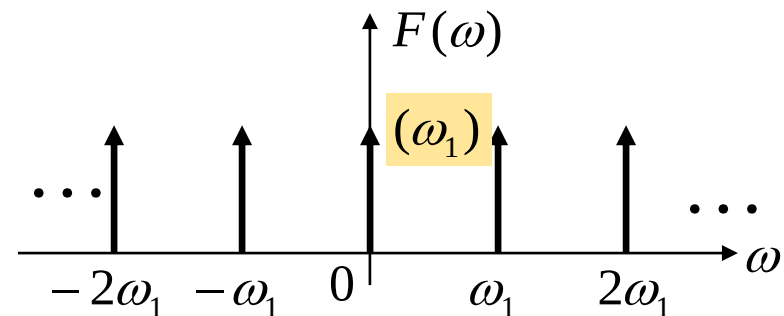
3.9 Fourier Transform of Periodic Signals



Fourier Series and Fourier Transform of the Periodic Unit Impulse Sequence:



$\times 2\pi$



$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_1)$$

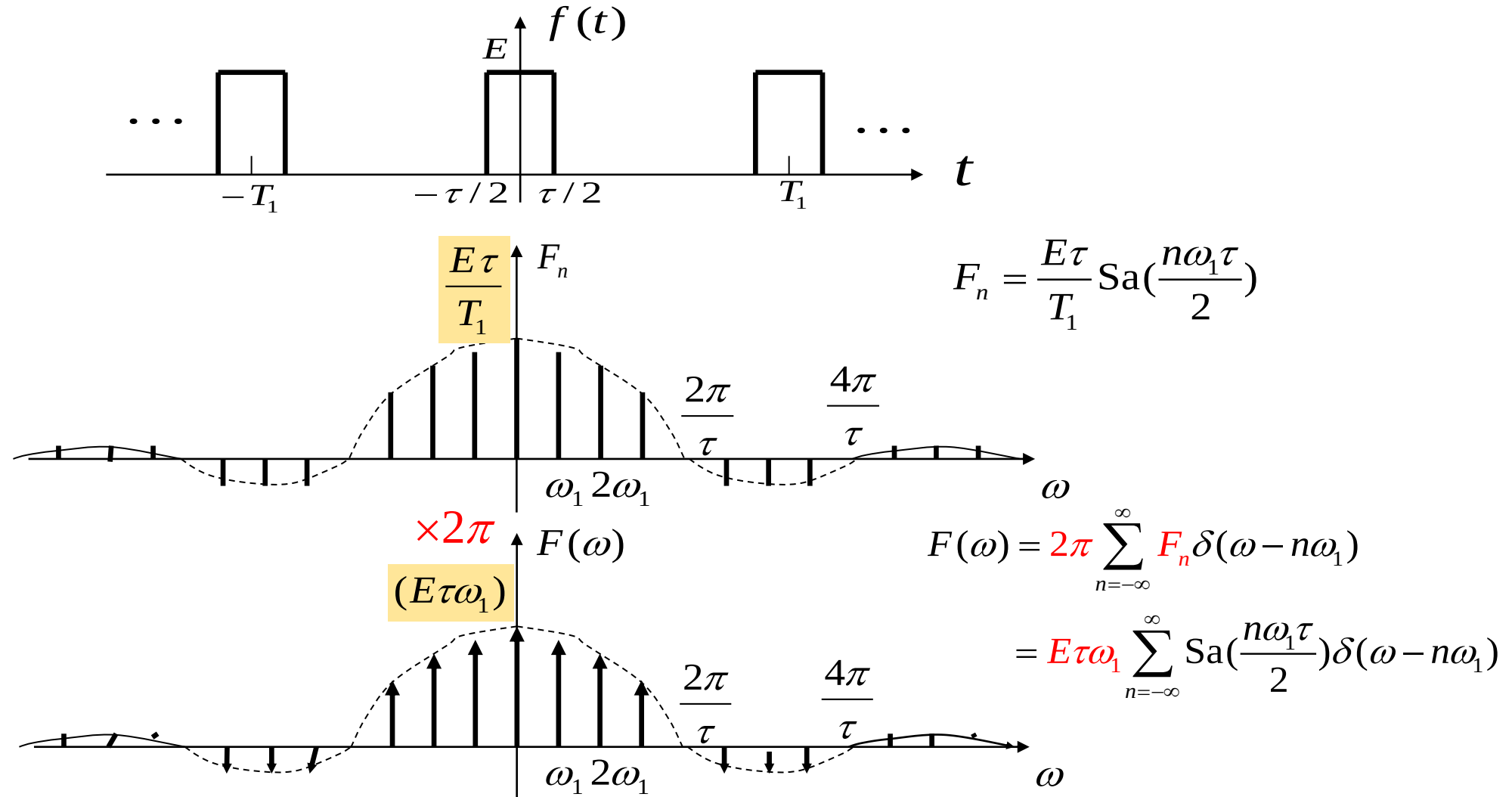
$$F_n = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} \delta(t) e^{-jn\omega_1 t} dt = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} \delta(t) dt = \frac{1}{T_1}$$

$$F(\omega) = \frac{2\pi}{T_1} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_1) = \omega_1 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_1)$$

3.9 Fourier Transform of Periodic Signals



Fourier Series and Fourier Transform of the Periodic Rectangular Pulse Signal:

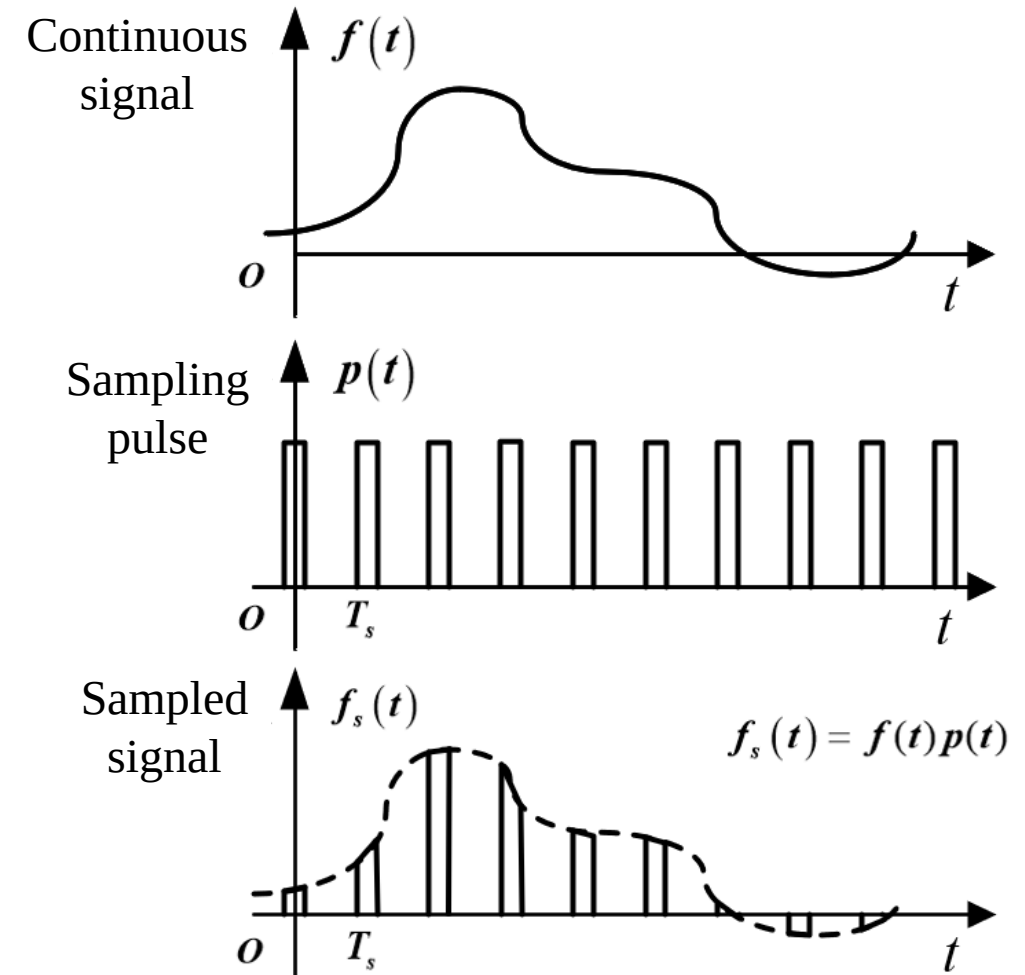
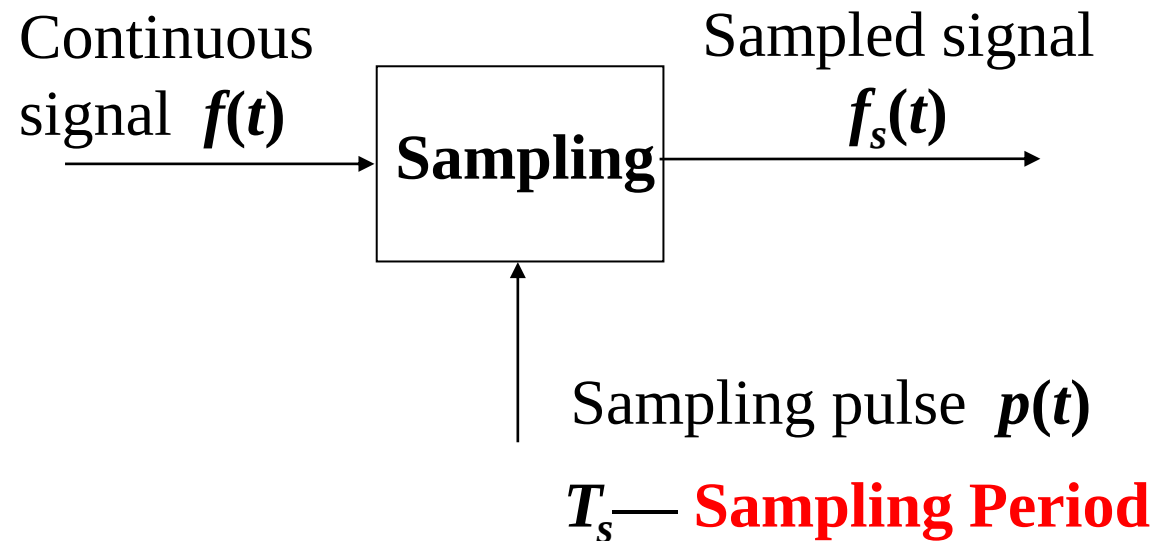


3.10 Fourier Transform of Sampled Signals

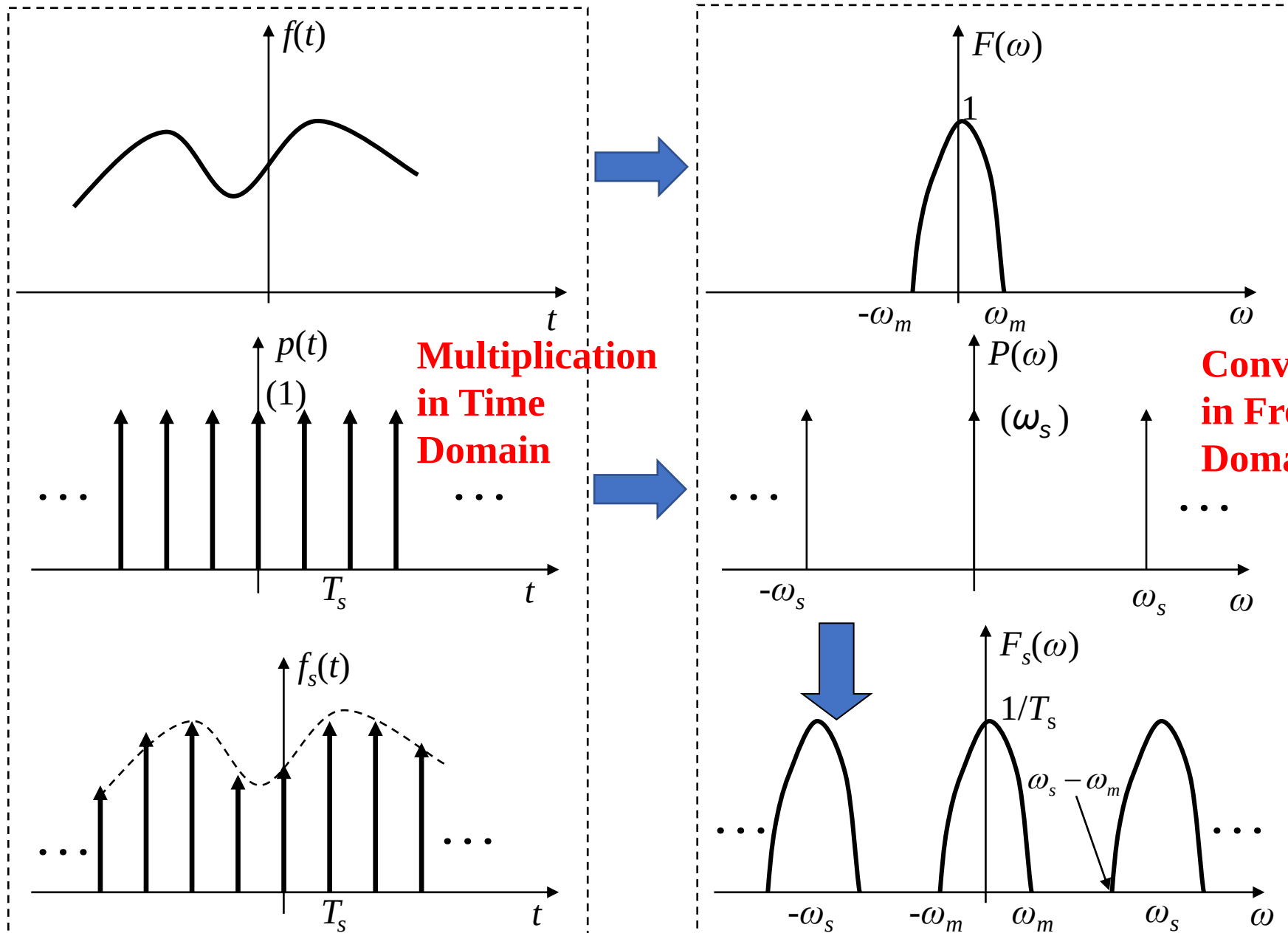
3.10.1 Sampling of Signals

Sampling -- Use a sampling pulse sequence to "sample" a series of discrete samples from a continuous signal.

This discrete signal is called a **sampled signal**.



3.10 Fourier Transform of Sampled Signals



Periodically extended in the frequency domain with period ω_s

$$F_s(\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} F(\omega - n\omega_s)$$

3.11 Sampling Theorem

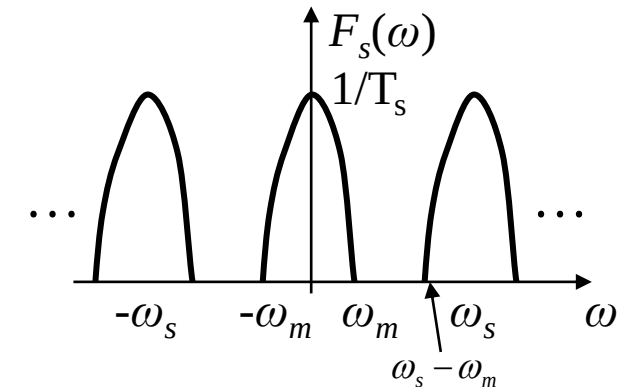
Time-domain Sampling Theorem: A band-limited signal $f(t)$, if its spectrum is confined to the range $-\omega_m \sim \omega_m$, can be uniquely represented by its equally spaced samples when the sampling interval is no larger than $\frac{1}{2f_m}$ (where $\omega_m = 2\pi f_m$) or the sampling frequency is greater than or equal to $2f_m$.

Nyquist Sampling Frequency: The minimum allowable sampling frequency.

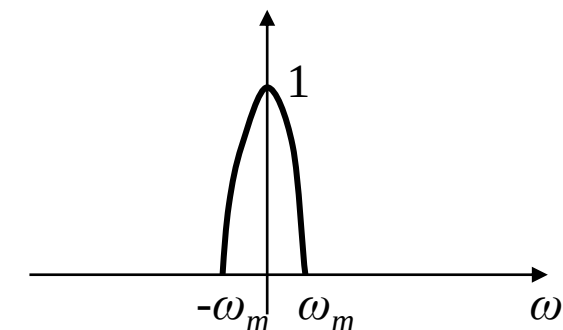
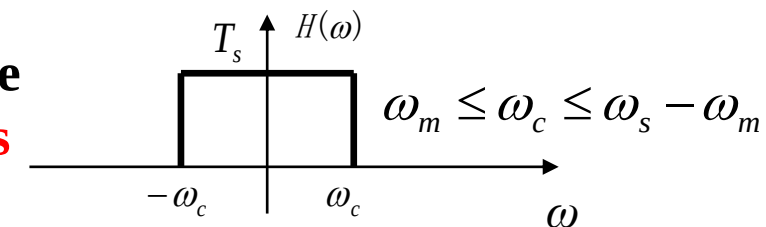
$$f_{s \min} = 2 f_m$$

Nyquist Sampling Interval: The maximum allowable sampling interval.

$$T_{s \max} = \frac{1}{f_{s \min}} = \frac{1}{2 f_m}$$



Frequency Response of the Ideal Low-pass Filter



Contents of This Lecture

4.1 Introduction

4.2 Definition of Laplace Transform

4.3 Laplace Transform of Typical Signals

4.4 Basic Properties of Laplace Transform

Learning Objectives of This Lecture

1. Understand the definition of **Laplace transform** and its relationship with Fourier transform;
2. Familiarize with the **basic properties** of Laplace transform.

From Fourier Transform to Laplace Transform

- **Fourier Transform** has a clear physical meaning.
- However, some signals do not satisfy the **Dirichlet conditions** (**sufficient but not necessary conditions for the existence of Fourier Transform**): $\int_{-\infty}^{\infty} |f(t)| dt < \infty$
- If we multiply by **an attenuation factor** $e^{-\sigma t}$ ($\sigma > 0$), then $f(t)e^{-\sigma t}$ converges and satisfies the Dirichlet conditions.

- $u(t)$



- $u(t)e^{-\sigma t}$

- Growing signal $e^{\alpha t}$ ($\alpha > 0$)



- $e^{\alpha t} e^{-\sigma t}$ ($\sigma > \alpha$)

- Periodic signal $\cos \omega_1 t$



- $e^{-\sigma t} \cos \omega_1 t$



Pierre Simon Laplace
(1749-1825)

4.2 Definition of Laplace Transform



Laplace Transform (LT)

For a signal $f(t)$, after multiplying by an attenuation factor $e^{-\sigma t}$ ($\sigma > 0$), it easily satisfies the absolute integrability condition. The new signal's Fourier transform is:

$$\begin{aligned} F_1(\omega) &= F[f(t) \cdot e^{-\sigma t}] = \int_{-\infty}^{+\infty} [f(t) e^{-\sigma t}] \cdot e^{-j\omega t} dt \\ &= \int_{-\infty}^{+\infty} f(t) \cdot e^{-(\sigma + j\omega)t} dt = F(\sigma + j\omega) \end{aligned}$$

Complex frequency: $s = \sigma + j\omega$

(Bilateral) Laplace Transform

$$F(s) = \int_{-\infty}^{\infty} f(t) e^{-st} dt$$

4.2 Definition of Laplace Transform

Unilateral Laplace Transform (practical signals are all **causal signals**. Thus, “unilateral” often does not appear in the name.)

$$F(s) = \int_{0_-}^{\infty} f(t) e^{-st} dt$$

$f(t) \xleftrightarrow{LT} F(s)$ $f(t)$ --the **original function**; $F(s)$ --the **image function**.

The unilateral Laplace Transform has included the effect of the initial condition:

$$F(s) = L[f(t)] = \int_{0_-}^{\infty} f(t) e^{-st} dt$$

Automatically includes the 0_- condition

$$L\left[\frac{df(t)}{dt}\right] = sF(s) - f(0_-)$$

Initial value, equal to 0 if there is no jump

Proof: $\int_{0_-}^{\infty} f'(t) e^{-st} dt = f(t) e^{-st} \Big|_0^{\infty} - \int_0^{\infty} f(t) (e^{-st})' dt$

Integration by parts

$$= f(\infty) e^{-s\infty} - f(0_-) + s \int_0^{\infty} f(t) e^{-st} dt$$

$$= sF(s) - f(0_-)$$

Given the signal $x(t) = e^{-\alpha t}u(t)$, its Laplace transform is ().

A $-\frac{1}{s + \alpha}$

B $\frac{\alpha}{s}$

C $e^{-\alpha s}$

☒ D $\frac{1}{s + \alpha}$

提交

4.2 Definition of Laplace Transform



Example 4-1: $x(t)=e^{-\alpha t}u(t)$, find its Laplace transform $X(s)$.

Solution: by definition

$$\begin{aligned} X(s) &= \int_{-\infty}^{\infty} x(t)e^{-st} dt = \int_{-\infty}^{\infty} e^{-\alpha t} u(t) e^{-st} dt \\ &= \int_0^{\infty} e^{-\alpha t} e^{-st} dt = \int_0^{\infty} e^{-(\alpha+s)t} dt = \int_0^{\infty} e^{-(\alpha+\sigma)t} e^{-j\omega t} dt \end{aligned}$$

when $\mathbf{Re\{\alpha+s\}=\alpha+\sigma>0}$, that is $\sigma > -\alpha$, the above integral is integrable

$$X(s) = \frac{1}{-(\alpha+s)} e^{-(\alpha+s)t} \Big|_0^{\infty} = \frac{1}{(\alpha+s)}, \quad \sigma > -\alpha$$

So,

$$e^{-\alpha t} u(t) \xleftrightarrow{LT} \frac{1}{s+\alpha}, \quad \text{Re}\{s\} > -\alpha$$

4.2 Definition of Laplace Transform



FT:
$$\begin{cases} F(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \\ f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{j\omega t} d\omega \end{cases}$$

The real frequency ω is the **oscillation frequency**.

LT:
$$\begin{cases} F(s) = \mathcal{L}[f(t)] = \int_{0_-}^{\infty} f(t) e^{-st} dt \\ f(t) = \mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s) e^{st} ds \end{cases}$$

The complex frequency $s = \sigma + j\omega$, where ω is the **oscillation frequency** and σ controls the **attenuation speed**.

4.2 Definition of Laplace Transform



Fourier Transform vs. Laplace Transform

Fourier Transform	Laplace Transform
The independent variable ω is a real number with a clear physical meaning – frequency .	The independent variable $s=\sigma+j\omega$ is a complex number , its physical meaning is unclear and it is usually called complex frequency .
The Fourier transform of a signal reflects the amplitude and initial phase of different frequency components, i.e., the spectrum of the signal.	The Laplace transform of a signal has no clear physical meaning .
The Fourier transform of the system's unit impulse response is called the system's frequency response , which represents the characteristic that the amplitude and phase of the system output change with the input frequency when cosine signals of different frequencies act on the system.	The Laplace transform of the system's unit impulse response is called the system's system function . Although it is abstract, it plays an important role in characterizing system properties and system analysis.
Mainly applied to frequency analysis of signals and systems, such as spectrum analysis of modulation, filtering, sampling, etc.	Mainly applied to solving differential equations, analysis of system functions and their zero-pole plots , etc.

4.3 Laplace Transform of Typical Signals



1. Step Function

$$L[u(t)] = \int_0^{\infty} 1 \cdot e^{-st} dt = \frac{1}{-s} e^{-st} \Big|_0^{\infty} = \frac{1}{s}$$

2. Single-sided Exponential Function

$$L[e^{-\alpha t} u(t)] = \int_0^{\infty} e^{-\alpha t} e^{-st} dt = \frac{e^{-(\alpha+s)t}}{-(\alpha+s)} \Big|_0^{\infty} = \frac{1}{s+\alpha} \quad (\sigma > -\alpha)$$

3. Unit Impulse Signal

$$L[\delta(t)] = \int_0^{\infty} \delta(t) \cdot e^{-st} dt = 1$$

$$L[\delta(t-t_0)] = \int_0^{\infty} \delta(t-t_0) \cdot e^{-st} dt = e^{-st_0}$$

4.3 Laplace Transform of Typical Signals



4. $t^n u(t)$

$$L[t^n] = \int_0^{\infty} t^n \cdot e^{-st} dt = \left. \frac{t^n}{-s} e^{-st} \right|_0^{\infty} + \frac{n}{s} \int_0^{\infty} t^{n-1} e^{-st} dt = \frac{n}{s} \int_0^{\infty} t^{n-1} e^{-st} dt$$

$$\text{so } L[t^n] = \frac{n}{s} L[t^{n-1}]$$

$$n=1 \quad L[t] = \frac{1}{s} L[t^0 u(t)] = \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2}$$

$$n=2 \quad L[t^2] = \frac{2}{s} L[t] = \frac{2}{s} \cdot \frac{1}{s^2} = \frac{2}{s^3}$$

$$n=3 \quad L[t^3] = \frac{3}{s} L[t^2] = \frac{3}{s} \cdot \frac{2}{s^3} = \frac{6}{s^4} \quad \dots$$

$$\text{so } L[t^n] = \frac{n!}{s^{n+1}}$$

4.4 Basic Properties of Laplace Transform



Note: The properties refer to those of the unilateral Laplace transform and only apply to **causal signals**, i.e., signals $f(t)$ that satisfy $f(t) = f(t)u(t)$.

1. Linearity

$$\text{if } f_i(t) \xleftrightarrow{LT} F_i(s)$$

$$\text{then } f(t) = \sum_{i=1}^N a_i f_i(t) \xleftrightarrow{LT} F(s) = \sum_{i=1}^N a_i F_i(s)$$

Example 4-2: Use the linearity property to find the Laplace transform of $x(t) = e^{-t}u(t) - e^{-2t}u(t)$.

$$\text{Solution: } e^{-t}u(t) \xleftrightarrow{LT} \frac{1}{s+1} \quad e^{-2t}u(t) \xleftrightarrow{LT} \frac{1}{s+2}$$

$$\text{So } x(t) \xleftrightarrow{LT} \frac{1}{s+1} - \frac{1}{s+2} = \frac{1}{(s+1)(s+2)}$$

4.4 Basic Properties of Laplace Transform



Example 4-3: Use the properties of Laplace transform to find the Laplace transform of $\cos(\omega_0 t)u(t)$.

Solution:

$$\cos(\omega_0 t)u(t) = \frac{1}{2} \left(e^{j\omega_0 t} + e^{-j\omega_0 t} \right) u(t)$$

$$e^{j\omega_0 t} u(t) \xleftrightarrow{LT} \frac{1}{s - j\omega_0}$$

$$e^{-j\omega_0 t} u(t) \xleftrightarrow{LT} \frac{1}{s + j\omega_0}$$

$$\therefore \cos(\omega_0 t)u(t) \xleftrightarrow{LT} \frac{s}{s^2 + \omega_0^2}$$

4.4 Basic Properties of Laplace Transform



2. Time Delay Property (Time-domain Shift Property)

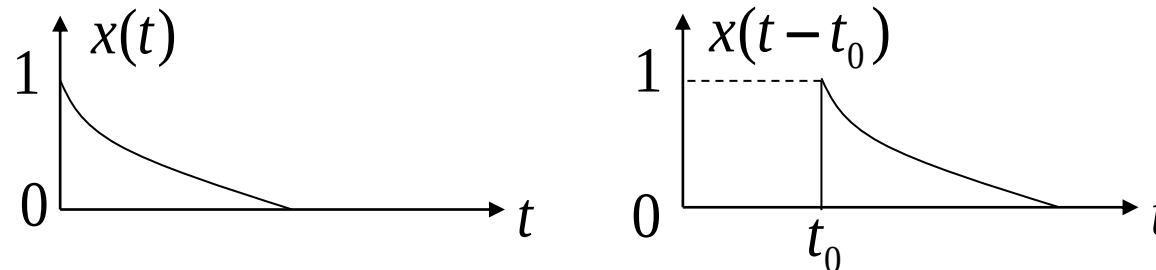
$$\text{Let } f(t) \xleftrightarrow{LT} F(s)$$

$$f(t - t_0)u(t - t_0) \xleftrightarrow{LT} F(s)e^{-st_0}$$

Note: This property cannot be used if the time-shifted signal is non-causal!

Under the definition of the **unilateral Laplace transform**,

The time delay of $e^{-\alpha t}u(t)$ is $e^{-\alpha(t-t_0)}u(t-t_0)$



The Laplace transform $F(s)$ of the signal $f(t) = u(t) - u(t - 1)$ is ()

☐ A $F(s) = \frac{1 + e^{-s}}{s}$

☒ B $F(s) = \frac{1 - e^{-s}}{s}$

☐ C $F(s) = \frac{1 - s}{s}$

☐ D $F(s) = \frac{1 - s}{e^{-s}}$

提交

4.4 Basic Properties of Laplace Transform



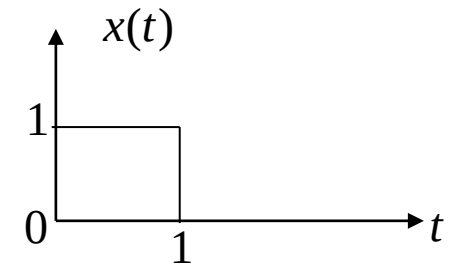
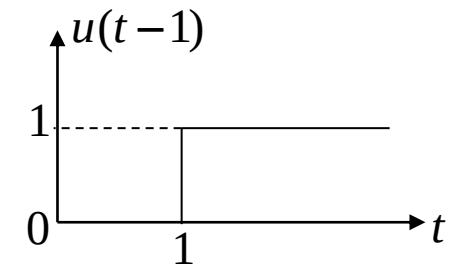
Example 4-4: Find the Laplace transform of $x(t) = u(t) - u(t-1)$.

Solution:

$$u(t) \xleftrightarrow{LT} \frac{1}{s}$$

$$u(t-1) \xleftrightarrow{LT} \frac{1}{s} e^{-s}$$

$$\text{so } x(t) = u(t) - u(t-1) \xleftrightarrow{LT} \frac{1 - e^{-s}}{s}$$



4.4 Basic Properties of Laplace Transform



3. s-domain Shift Property (Frequency Shift Property)

$$\text{Let} \quad f(t) \xleftrightarrow{LT} F(s)$$

$$\text{Then} \quad f(t)e^{s_0 t} \xleftrightarrow{LT} F(s - s_0)$$

Example 4-5: Use the s-domain shift property to find the Laplace transforms of $e^{-\alpha t}u(t)$ and $te^{-\alpha t}u(t)$.

Solution:

$$u(t) \xleftrightarrow{LT} \frac{1}{s}$$

$$tu(t) \xleftrightarrow{LT} \frac{1}{s^2}$$

$$e^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{s + \alpha}$$

$$te^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{(s + \alpha)^2}$$

4.4 Basic Properties of Laplace Transform



4. Time-domain Scaling Property

$$\text{if } f(t) \xleftrightarrow{LT} F(s)$$

$$\text{then } f(at) \xleftrightarrow{LT} \frac{1}{a} F\left(\frac{s}{a}\right)$$

The constant $a > 0$, otherwise the Laplace transform is meaningless.

Example 4-6: Use the time-domain scaling property to find the Laplace transform of $e^{-\alpha t}u(t)$.

Solution:

$$e^{-t}u(t) \xleftrightarrow{LT} \frac{1}{s+1}$$

$$\text{thus, } e^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{\alpha} \frac{1}{\frac{s}{\alpha} + 1} = \frac{1}{\alpha} \frac{\alpha}{s + \alpha} = \frac{1}{s + \alpha}$$

4.4 Basic Properties of Laplace Transform



$$\text{if } f(t)u(t) \xleftrightarrow{LT} F(s)$$

When there are both time delay and time-domain scaling,

$$f(at - b)u(at - b) \xleftrightarrow{LT} ? \quad (a > 0, b > 0)$$

First perform time delay, then time-domain scaling:

$$f(t - b)u(t - b) \xleftrightarrow{LT} F(s)e^{-bs}$$

$$f(at - b)u(at - b) \xleftrightarrow{LT} \frac{1}{a} F\left(\frac{s}{a}\right) e^{-\frac{bs}{a}}$$

It is also OK to perform time-domain scaling first, then time delay.

4.4 Basic Properties of Laplace Transform



Example 4-7: Find the Laplace Transform of $\delta(3t - 2)$.

Solution 1: First Time Delay, Then Time-domain Compression

From the time-shift property $f(t - t_0)u(t - t_0) \xleftrightarrow{LT} F(s)e^{-st_0}$ and $\delta(t) \xleftrightarrow{LT} 1$

we have: $\delta(t - 2) \xleftrightarrow{LT} e^{-2s}$

From the time-domain scaling property $f(at) \xleftrightarrow{LT} \frac{1}{a} F\left(\frac{s}{a}\right)$, we get: $\delta(3t - 2) \xleftrightarrow{LT} \frac{1}{3} e^{-\frac{2}{3}s}$

Solution 2: First Time-domain Compression, Then Time Delay

From the time-domain scaling property $f(at) \xleftrightarrow{LT} \frac{1}{a} F\left(\frac{s}{a}\right)$, we get: $\delta(3t) \xleftrightarrow{LT} \frac{1}{3}$

From the time-shift property $f(t - t_0)u(t - t_0) \xleftrightarrow{LT} F(s)e^{-st_0}$:

$$\delta(3t - 2) = \delta\left[3\left(t - \frac{2}{3}\right)\right] \xleftrightarrow{LT} \frac{1}{3} e^{-\frac{2}{3}s}$$

5. Time-domain Differentiation Property

Let $f(t) \xleftrightarrow{LT} F(s)$

$$f'(t) \xleftrightarrow{LT} sF(s) - f(0_-)$$

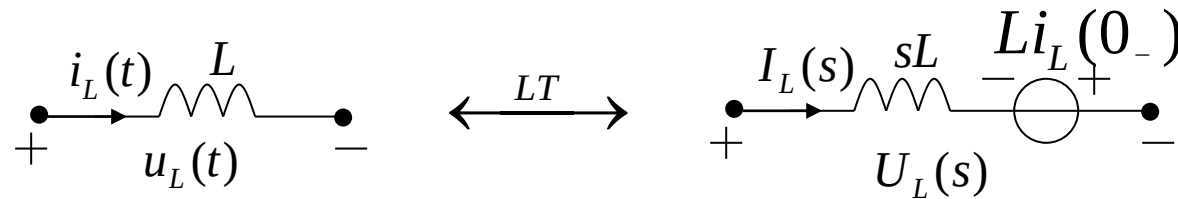
$$f^{(n)}(t) \xleftrightarrow{LT} s^n F(s) - \sum_{i=0}^{n-1} s^{n-i-1} f^{(i)}(0_-)$$

With zero state, differentiating once in the time domain corresponds to multiplying by s in the frequency domain.

Example 4-8: Find the Laplace Transform of inductor voltage $u_L(t)$.

Solution:

$$u_L(t) = L \frac{di_L(t)}{dt} \xleftrightarrow{LT} U_L(s) = sLI_L(s) - Li_L(0_-)$$

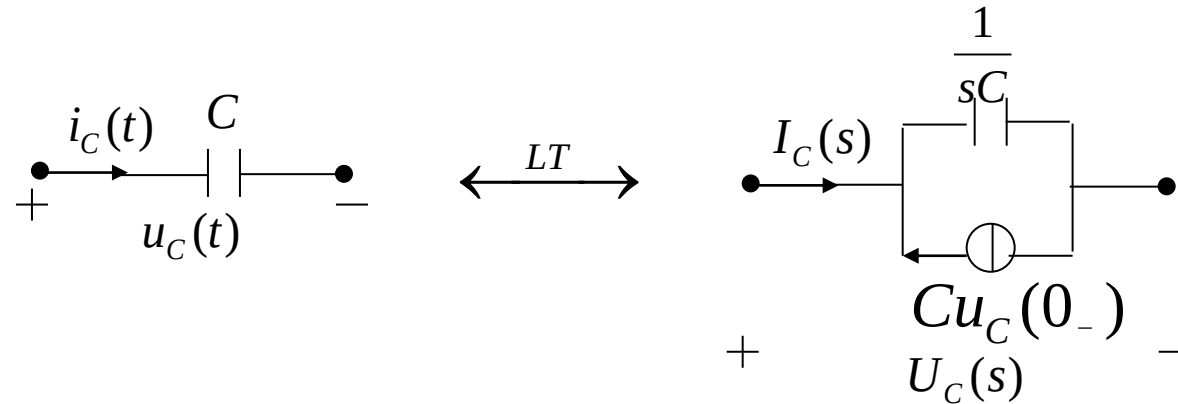


4.4 Basic Properties of Laplace Transform



Similarly, for **the current in a capacitor**,

$$i_C(t) = C \frac{du_C(t)}{dt} \xleftrightarrow{LT} I_C(s) = sCU_C(s) - Cu_C(0_-)$$



4.4 Basic Properties of Laplace Transform



6. Time-domain Integration Property

$$f(t) \xleftrightarrow{LT} F(s)$$

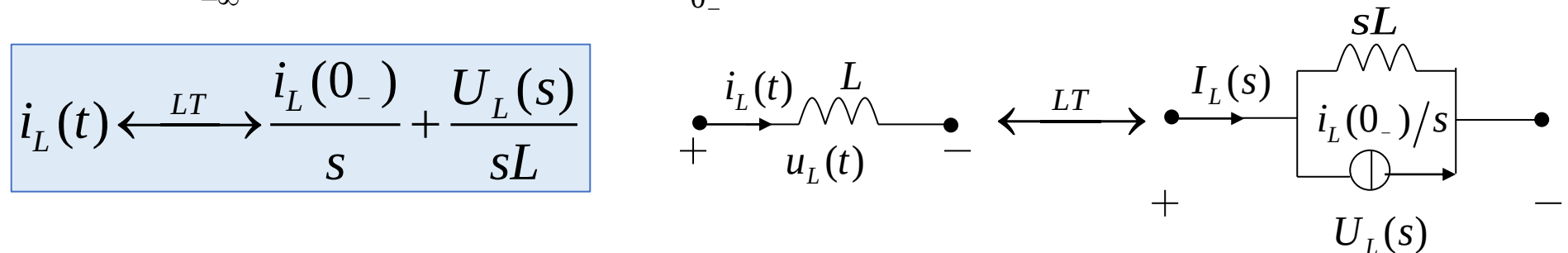
then $y(t) = \int_{-\infty}^t f(\tau) d\tau = \int_{-\infty}^{0_-} f(\tau) d\tau + \int_{0_-}^t f(\tau) d\tau = y(0_-) + \int_{0_-}^t f(\tau) d\tau$

$$y(t) = \int_{-\infty}^t f(\tau) d\tau \xleftrightarrow{LT} \frac{y(0_-)}{s} + \frac{F(s)}{s}$$

Example 4-9: Find the Laplace transform of the current $i_L(t)$ in an inductor.

Solution: $i_L(t) = \frac{1}{L} \int_{-\infty}^t u_L(\tau) d\tau = i_L(0_-) + \frac{1}{L} \int_{0_-}^t u_L(\tau) d\tau$

thus,



4.4 Basic Properties of Laplace Transform

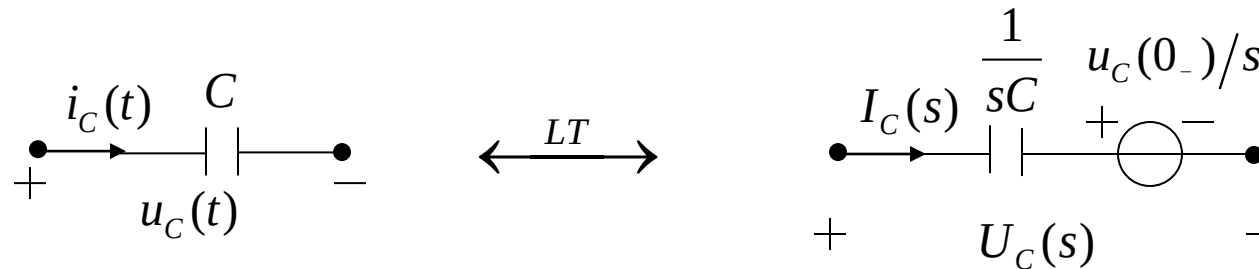


Similarly, **the voltage across a capacitor**

$$u_C(t) = \frac{1}{C} \int_{-\infty}^t i_C(\tau) d\tau = u_C(0_-) + \frac{1}{C} \int_{0_-}^t i_C(\tau) d\tau$$

Its Laplace transform is

$$u_C(t) \xleftrightarrow{LT} \frac{u_C(0_-)}{s} + \frac{I_C(s)}{sC}$$



The **time-domain calculus properties** of Laplace transform are quite convenient when **analyzing the transient state of circuits**.

7. s-domain Differentiation Property

Let $f(t) \xleftrightarrow{LT} F(s)$

s-domain differentiation property:

$$-tf(t) \xleftrightarrow{LT} \frac{dF(s)}{ds} \quad tf(t) \xleftrightarrow{LT} -\frac{dF(s)}{ds}$$

Example 4-10: Given $e^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{s+\alpha}$, use the s-domain differentiation property to find the Laplace transform of $te^{-\alpha t}u(t)$.

Solution:
$$\frac{d}{ds} \left[\frac{1}{s+\alpha} \right] = \frac{-1}{(s+\alpha)^2}$$

Thus,
$$te^{-\alpha t}u(t) \xleftrightarrow{LT} \frac{1}{(s+\alpha)^2}$$

8. s-domain Integration Property

$$\frac{f(t)}{t} \xleftrightarrow{LT} \int_s^{\infty} F(\lambda) d\lambda$$

Example 4-11: Given $(1 - e^{-\alpha t})u(t) \xleftrightarrow{LT} \frac{1}{s} - \frac{1}{s + \alpha}$, find the Laplace transform of $\frac{1 - e^{-\alpha t}}{t} u(t)$.

Solution:

$$\begin{aligned} \int_s^{\infty} \left(\frac{1}{\lambda} - \frac{1}{\lambda + \alpha} \right) d\lambda &= \ln \lambda \Big|_s^{\infty} - \ln(\lambda + \alpha) \Big|_s^{\infty} \\ &= -\ln s + \ln(s + \alpha) \\ &= -\ln \frac{s}{s + \alpha} \end{aligned}$$

Thus,
$$\frac{1 - e^{-\alpha t}}{t} u(t) \xleftrightarrow{LT} -\ln \frac{s}{s + \alpha}$$

9. Convolution Theorem (Time-domain, Frequency-domain)

Let $f_1(t) \xleftrightarrow{LT} F_1(s), f_2(t) \xleftrightarrow{LT} F_2(s)$

then $f_1(t) * f_2(t) \xleftrightarrow{LT} F_1(s) \cdot F_2(s)$

$$f_1(t) f_2(t) \xleftrightarrow{LT} \frac{1}{2\pi j} F_1(s) * F_2(s)$$

This theorem can be proved using the definition of Laplace transform and the time-delay property.

Example 4-12: Given the unit impulse response and input of a system as follows, find the zero-state response $y(t)$.

$$h(t) = e^{-\alpha t} u(t), \quad x(t) = u(t)$$

Solution: $h(t) = e^{-\alpha t} u(t) \xleftrightarrow{LT} \frac{1}{s + \alpha} \quad x(t) = u(t) \xleftrightarrow{LT} \frac{1}{s}$

so $y(t) = x(t) * h(t) \xleftrightarrow{LT} \frac{1}{s(s + \alpha)} = \frac{1}{\alpha} \left(\frac{1}{s} - \frac{1}{s + \alpha} \right)$

$$y(t) = \frac{1}{\alpha} (1 - e^{-\alpha t}) u(t)$$

10. Initial-Value and Final-Value Theorems

Suppose the signal $f(t)$ is **causal**, and the Laplace transforms of $f(t)$ and its derivatives exist, then

(1) Initial-Value Theorem:

$$\lim_{t \rightarrow 0_+} f(t) = f(0_+) = \lim_{s \rightarrow \infty} sF(s)$$

It represents the high-frequency component ($\omega \rightarrow \infty$), indicating the mutation of the signal at $t=0$.

If $f(t)$ contains an impulse function $k\delta(t)$,

$$F(s) = k + F_1(s), \quad F_1(s) \text{ is a proper fraction}$$

$$\lim_{t \rightarrow 0_+} f(t) = \lim_{s \rightarrow \infty} [sF(s) - ks] = \lim_{s \rightarrow \infty} sF_1(s)$$

(2) Final-Value Theorem: if $f(\infty)$ exists, then

$$\lim_{t \rightarrow \infty} f(t) = f(\infty) = \lim_{s \rightarrow 0} sF(s)$$

It represents the DC component ($\omega \rightarrow 0$), obtaining the final value of the circuit in steady state.

When circuits and other systems are relatively complex, the initial value and final value of the function can be directly obtained without performing the inverse Laplace transform.

4.4 Basic Properties of Laplace Transform



Example 4-13: Given $u(t) \xleftrightarrow{LT} \frac{1}{s}$, use the properties of Laplace transform to find the initial value and final value of $u(t)$ and $\cos(\omega_0 t)u(t)$ respectively.

Solution:

$$u(t) \big|_{t=0_+} = u(t) \big|_{t=\infty} = s \cdot \frac{1}{s} = 1$$

$$\cos(\omega_0 t)u(t) \xleftrightarrow{LT} \frac{s}{s^2 + \omega_0^2} \qquad \cos(\omega_0 t) \big|_{t=0_+} = \lim_{s \rightarrow \infty} s \frac{s}{s^2 + \omega_0^2} = 1$$

$\cos(\omega_0 t)u(t)$ is a constant-amplitude oscillating function (the roots of the denominator of $sF(s)$ are conjugate symmetric on the imaginary axis), so its final value does not exist.

4.4 Basic Properties of Laplace Transform



Example 4-14: Use the properties of Laplace transform to find the Laplace transform of $te^{-\alpha t} \cos(\omega_0 t)u(t)$.

Solution: Apply the **Frequency-Shifting Property** and the **s-domain Differentiation Property**.

$$\begin{aligned} e^{-\alpha t} \cos(\omega_0 t)u(t) &\xleftrightarrow{LT} \frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2} \\ -te^{-\alpha t} \cos(\omega_0 t)u(t) &\xleftrightarrow{LT} \frac{d}{ds} \left[\frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2} \right] \\ \frac{d}{ds} \left[\frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2} \right] &= \frac{\omega_0^2 - (s + \alpha)^2}{[(s + \alpha)^2 + \omega_0^2]^2} \\ \therefore te^{-\alpha t} \cos(\omega_0 t)u(t) &\xleftrightarrow{LT} \frac{(s + \alpha)^2 - \omega_0^2}{[(s + \alpha)^2 + \omega_0^2]^2} \end{aligned}$$

4.4 Basic Properties of Laplace Transform



Summary

Property	Time-domain	Complex Frequency Domain
Linearity	$\sum_{i=1}^N a_i f_i(t)$	$\sum_{i=1}^N a_i F_i(s)$
Time-Delay	$f(t - t_0)u(t - t_0)$	$e^{-st_0} F(s)$
Frequency-Shifting	$f(t)e^{-at}$	$F(s + a)$
Time-Scaling	$f(at) \quad (a > 0)$	$\frac{1}{a} F\left(\frac{s}{a}\right)$
Time-Domain Convolution Theorem	$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$
Frequency-Domain Convolution Theorem	$f_1(t) \cdot f_2(t)$	$F_1(s) * F_2(s) / (2\pi j)$

Note: The above properties only apply to **causal signals**, i.e., signals $f(t)$ that satisfy $f(t) = f(t)u(t)$.

4.4 Basic Properties of Laplace Transform



Summary

Property	Time-domain	Complex Frequency Domain
Time-domain Differentiation	$\frac{df(t)}{dt}$	$sF(s) - f(0^-)$
Time-domain Integration	$\int_{-\infty}^t f(\tau) d\tau$	$\frac{F(s)}{s} + \frac{f^{-1}(0^-)}{s}$
s-domain Differentiation	$-tf(t)$	$\frac{dF(s)}{ds}$
s-domain Integration	$f(t)/t$	$\int_s^{\infty} F(\lambda) d\lambda$
Initial Value Theorem	$\lim_{t \rightarrow 0^+} f(t) = f(0^+) = \lim_{s \rightarrow \infty} sF(s)$	
Final Value Theorem	$\lim_{t \rightarrow \infty} f(t) = f(\infty) = \lim_{s \rightarrow 0} sF(s)$	

Note: The above properties only apply to **causal signals**, i.e., signals $f(t)$ that satisfy $f(t) = f(t)u(t)$.

- Repeat the examples in the lecture notes.
- In the textbook:
 - 6.29 (b) $x(t) = tu(t) * \cos(2\pi t)u(t)$
(c) $x(t) = \int_0^t e^{-3\tau} \cos(2\tau) d\tau$
 - 6.35(a)
 - 6.36(b)